1 Testing watertight spaces not intended to hold liquids and car go holds intended to hold ballast by filling them with water is not compulsory. When testing by filling with water is not carried out, a hose test shall be ca rried out where practicable. This test shall be carried out in the most advanced stage of the fitting out of the ship. Where a hose test

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SOURCE: MSC.194(80).pdf
water is not carried out, a hose test shall be ca rried out where practicable. This test shall be carried out in the most advanced stage of the fitting out of the ship. Where a hose test is not practicable because of possible dama ge to machinery, electrical equipment insulation or outfitting items, it may be replaced by a careful visual examination of welded connections, supported where deemed necessary by means such as a dye penetrant test or an ultrasonic leak test or an equivalent test. In any case a thorough inspection of the watertight bulkheads shall be carried out.

2 The forepeak, double bottom (including duct keels) and inner skins shall be tested with water to a head corresponding to the requirements of regulation 10.1.

RESOLUTION MSC.194(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED-30 -

3 Tanks which are intended to hold liquids, and which form part of the watertight subdivision of the ship, shall be tested for tightness and structural strength with water to a head corresponding to its design pressure. The water head is in no case to be less than the

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SOURCE: MSC.194(80).pdf
subdivision of the ship, shall be tested for tightness and structural strength with water to a head corresponding to its design pressure. The water head is in no case to be less than the top of the air pipes or to a level of 2.4 m above the top of the tank, whichever is the greater.

4 The tests referred to in paragraphs 2 and 3 are for the purpose of ensuring that the subdivision structural arrangements are watertight and are not to be regarded as a test of the fitness of any compartment for the storage of oil fuel or for ot her special purposes for which a test of a superior character may be required depending on the height to which the liquid has access in the tank or its connections.

Regulation 12
Peak and machinery space bulkheads, shaft tunnels, etc.

1 A collision bulkhead shall be fitted which shall be watertight up to the bulkhead deck. This bulkhead shall be located at a di stance from the forward perpendicular of not less than 0.05 L or 10 m, whichever is the less, and, except as may be permitted by the Administration, not more than 0.08L or 0.05L + 3 m, whichever is the greater.

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SOURCE: MSC.194(80).pdf
less than 0.05 L or 10 m, whichever is the less, and, except as may be permitted by the

Administration, not more than 0.08L or 0.05L + 3 m, whichever is the greater.

2 Where any part of the ship below the waterline extends forward of the forward perpendicular, e.g. a bulbous bow, the dist ances stipulated in paragraph 1 shall be measured from a point either:

.1 at the mid-length of such extension;

.2 at a distance 0.015 L forward of the forward perpendicular; or

.3 at a distance 3 m forward of the forward perpendicular,

whichever gives the smallest measurement.

3 The bulkhead may have steps or recesse s provided they are within the limits prescribed in paragraph 1 or 2.

4 No doors, manholes, access openings, ve ntilation ducts or any other openings shall be fitted in the collision bulkhead below the bulkhead deck.

5.1 Except as provided in paragraph 5.2, th e collision bulkhead may be pierced below the bulkhead deck by not more than one pipe fo r dealing with fluid in the forepeak tank, provided that the pipe is fitted with a scre w-down valve capable of being operated from

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SOURCE: MSC.194(80).pdf
the bulkhead deck by not more than one pipe fo r dealing with fluid in the forepeak tank, provided that the pipe is fitted with a scre w-down valve capable of being operated from above the bulkhead deck, the valve chest be ing secured inside th e forepeak to the collision bulkhead. The Administration may, however, authorize the fitting of this valve on the after side of the collision bulkhead provided that the valve is readily accessible under all service conditions and the space in wh ich it is located is not a cargo space. All valves shall be of steel, bron ze or other approved ductile ma terial. Valves of ordinary cast iron or similar material are not acceptable.

5.2 If the forepeak is divided to hold two different kinds of liquids the Administration may allow the collision bulkhead to be pier ced below the bulkhead deck by two pipes, each of which is fitted as required by paragraph 5.1, provided the Administration is satisfied that there is no practical alternative to the fitting of such a second pipe and that, having regard to the additional subdivision provided in the forepeak, the safety of the ship is maintained.
RESOLUTION MSC.194(80)

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SOURCE: MSC.194(80).pdf
satisfied that there is no practical alternative to the fitting of such a second pipe and that, having regard to the additional subdivision provided in the forepeak, the safety of the ship is maintained.
RESOLUTION MSC.194(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED-

6 Where a long forward superstructure is fitted the collision bulkhead shall be extended weathertight to the deck next above the bulkhead deck. The extension need not be fitted directly above the bulkhead below provided it is located within the limits prescribed in paragraph 1 or 2 with the exception permitted by paragraph 7 and that the part of the deck which forms the step is ma de effectively weathertight. The extension shall be so arranged as to preclude the possibility of the bow door causing damage to it in the case of damage to, or detachment of, a bow door.

7 Where bow doors are fitted and a sloping lo ading ramp forms part of the extension of the collision bulkhead above the bulkhead deck the ramp shall be weathertight over its

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SOURCE: MSC.194(80).pdf
the case of damage to, or detachment of, a bow door.

7 Where bow doors are fitted and a sloping lo ading ramp forms part of the extension of the collision bulkhead above the bulkhead deck the ramp shall be weathertight over its complete length. In cargo ships the part of the ramp which is more than 2.3 m above the bulkhead deck may extend forward of the limit specified in paragraph 1 or 2. Ramps not meeting the above requirements shall be disr egarded as an extension of the collision bulkhead.

8 The number of openings in the extensi on of the collision bulkhead above the freeboard deck shall be restricted to the mi nimum compatible with the design and normal operation of the ship. All such openings shall be capable of being closed weathertight.

9 Bulkheads shall be fitted separating the machinery space from cargo and accommodation spaces forward and aft and made watertight up to the bulkhead deck. In passenger ships an afterpeak bulkhead shall also be fitted and made watertight up to the bulkhead deck. The afterpeak bulkhead may, however, be stepped below the bulkhead deck, provided the degree of safety of the ship as regards subdivision is not thereby

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SOURCE: MSC.194(80).pdf
bulkhead deck. The afterpeak bulkhead may, however, be stepped below the bulkhead deck, provided the degree of safety of the ship as regards subdivision is not thereby diminished.

10 In all cases stern tubes shall be enclosed in watertight spaces of moderate volume. In passenger ships the stern gland shall be situ ated in a watertight shaft tunnel or other watertight space separate from the stern tube compartment and of such volume that, if flooded by leakage through the stern gland, the bulkhead deck will not be immersed. In cargo ships other measures to minimize the dange r of water penetrating into the ship in case of damage to stern tube arrangement s may be taken at the discretion of the Administration.

Regulation 13
Openings in watertight bulkheads below the bulkhead deck in passenger ships

1 The number of openings in watertight bul kheads shall be reduced to the minimum compatible with the design and proper working of the ship, satisfactory means shall be

provided for closing these openings.

2.1 Where pipes, scuppers, electric cables , etc., are carried through watertight

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SOURCE: MSC.194(80).pdf
compatible with the design and proper working of the ship, satisfactory means shall be provided for closing these openings.

2.1 Where pipes, scuppers, electric cables , etc., are carried through watertight bulkheads, arrangements shall be made to ensure the watertight integrity of the bulkheads.

2.2 Valves not forming part of a piping sy stem shall not be permitted in watertight bulkheads.

2.3 Lead or other heat sensitive materials sh all not be used in systems which penetrate watertight bulkheads, where deterioration of such systems in the event of fire would impair the watertight integrity of the bulkheads.

3 No doors, manholes, or access openings ar e permitted in watertight transverse bulkheads dividing a cargo space from an adjoining cargo space, except as provided in paragraph 9.1 and in regulation 14.
RESOLUTION MSC.194(80)
(adopted on 20 May 2005)